RAJALAKSHMI ENGINEERING COLLEGE

Rohan K 251001216-2025-IT

R2

Started on	Wednesday, 12 August 2026, 9:21 AM
State	Finished
Completed on	Wednesday, 12 August 2026, 9:34 AM
Time taken	12 mins 30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    int i = 1;
    int s = 1;
    int count = 0;

    scanf("%d", &n);

    while (s <= n)
    {
        i++;
        s += i;

        count += 3;
    }

    count += 3;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Rohan K 251001216-2025-IT

R2

Started on	Wednesday, 12 August 2026, 9:35 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 7:53 PM
Time taken	13 days 10 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
/*#include <stdio.h>

void func(int n)
{
    int count=0;

    if (n == 1)
        count = 1;
    else
        count = 5 * n + 2;

    printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);

    func(n);

    return 0;
}*/
#include <stdio.h>

void func(int n)
{
    int count = 0;
```

```
29
30      if (n == 1)
31      {
32          count++;
33      }
34      else
35      {
36          for (int i = 1; i <= n; i++)
37          {
38              for (int j = 1; j <= 5; j++)
39              {
40
41
42                  count++;
43              }
44          }
45      }
46      count+=2;
47
48      printf("%d", count);
49  }
50
51  int main()
52  {
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Rohan K 251001216-2025-IT

R2

Started on	Wednesday, 19 August 2026, 8:36 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 7:54 PM
Time taken	6 days 11 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int num, i, count = 0;

    scanf("%d", &num);

    for (i = 1; i <= num; ++i) {
        count++;

        if (num % i == 0) {
            count++;
        }

        count++;
    }

    count++;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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R2

Started on	Wednesday, 19 August 2026, 8:53 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 7:54 PM
Time taken	6 days 11 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, i, j, k;
    int count = 0;

    scanf("%d", &n);

    count++;

    for (i = n / 2; i < n; i++)
    {
        count++;

        for (j = 1; j < n; j = 2 * j)
        {
            count++;

            for (k = 1; k < n; k = k * 2)
            {
                count++;
                count++;
            }

            count++;
        }

        count++;
    }

    count++;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Rohan K 251001216-2025-IT

R2

Started on	Wednesday, 19 August 2026, 9:07 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 7:55 PM
Time taken	6 days 10 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, rev = 0, remainder;
    int count = 0;

    scanf("%d", &n);

    count++;   // rev = 0

    while (n != 0)
    {
        count++;   // while condition

        remainder = n % 10;
        count++;   // remainder = n % 10

        rev = rev * 10 + remainder;
        count++;   // rev = rev * 10 + remainder

        n /= 10;
        count++;   // n /= 10
    }

    count++;       // final while condition

    count++;       // print(rev)

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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